THE KIDNEYS & URINARY SYSTEM

Your Body’s Natural Filtration System

Every day, your kidneys filter nearly **180 liters (about 47 gallons)** of blood, removing waste products, balancing minerals, regulating blood pressure, and maintaining the body’s fluid balance. They are among the hardest-working organs in the human body.

Because the kidneys continuously filter the bloodstream, scientists are studying whether they may also filter, retain, or eliminate microscopic plastic particles that enter our circulation.

Why the Kidneys Matter

Everything that enters your bloodstream eventually passes through your kidneys.

These remarkable organs act as the body’s natural filtration system, helping remove toxins and waste products while returning essential nutrients and water back into circulation.

If microplastics enter the bloodstream, researchers want to understand whether the kidneys can eliminate them, whether some particles become trapped within kidney tissue, or whether long-term exposure could affect kidney function.

What the Research Has Found

Scientists have detected microplastics in donated human kidney tissue, confirming that these particles can reach one of the body’s primary filtering organs.

Researchers believe the smallest particles, particularly nanoplastics, may be more capable of passing through biological barriers and interacting with kidney cells. Laboratory studies have suggested that these particles may trigger inflammation, oxidative stress, or cellular injury under certain conditions.

However, human research remains limited.

At present, there is **no definitive evidence** that microplastics directly cause chronic kidney disease, kidney failure, or reduced kidney function in humans.

Understanding how these particles move through and interact with the kidneys has become an important area of ongoing research.

What This Means

Your kidneys are constantly working to keep your body in balance.

The discovery of microplastics in kidney tissue confirms that these particles can reach organs involved in filtration and waste removal. What remains uncertain is whether they simply pass through the kidneys or whether long-term accumulation may have biological effects.

Researchers are now investigating whether repeated exposure over many years contributes to inflammation or changes in kidney function.

For now, the evidence highlights another way environmental exposures may interact with the body’s natural defense systems.

Key Takeaways

- The kidneys continuously filter the blood and help remove waste from the body.
- Scientists have detected microplastics in donated human kidney tissue.
- Researchers are investigating whether these particles influence kidney function or simply pass through the body’s filtration system.
- Laboratory studies suggest possible effects on kidney cells, but human evidence remains limited.
- There is currently **no definitive evidence** that microplastics directly cause chronic kidney disease, but this remains an important area of ongoing research.